Paper Summary: Stem Wood Quantity and Quality of Young Paulownia (Kiri) Planted in Tochigi, Japan, for Biofuel

This is a summary by Mitsue-it, not a copy of the paper. Please check the original for exact content. References such as "Original Table 5" point to the table or section in the original paper.

Original: Nezu I, Sawada N, Nihei M, Kato S, Aizawa M, Aruga K, Ohshima J, Yokota S, Ishiguri F (2026) *Quantity and Quality of Stem Wood Concerning Biofuel Production from Young Paulownia Species Planted in Tochigi, Japan.* BioEnergy Research 19:6. https://doi.org/10.1007/s12155-025-10943-8 Utsunomiya University and others. Funded by NEDO (JPNP21002). Licence: CC BY-NC-ND 4.0 (non-commercial sharing allowed; sharing adapted versions is not).

1. What was studied

- Paulownia was planted in July 2020 on abandoned farmland (0.89 ha) in Mashiko, Tochigi, at 3 × 3 m spacing. Fertilising, weeding, pest and disease control and pruning were done every year (Original, Materials and Methods).
- Fifteen 4-year-old trees (diameter at breast height 15.6 ± 1.2 cm, height 9.7 ± 0.8 m) were measured for stem volume, weight, moisture content, density, chemical composition, calorific value and elements.
- For comparison, calorific value and ash were also measured in nine Japanese species: sugi, hinoki, karamatsu, akamatsu, konara, yamazakura, kuri, hōnoki and Paulownia tomentosa (Original Tables 5, 6).

2. Main results

Item	Kiri (4 years)	Sugi	Original
Calorific value (wood, per dry kg)	18.9 MJ/kg	20.2 MJ/kg	Table 5
Basic density (dry weight / green volume)	209 kg/m³ (stem mean 255)	302 kg/m³	Tables 5, 1
Calorific value per m³ of wood	4.0 GJ/m³ (lowest of the 10 species)	6.1 GJ/m³	Table 5
Ash (wood)	1.7%	0.4%	Table 6
Ash (bark)	4.5%	0.5%	Table 6
Potassium	0.4% wood, 1.5% bark	not measured	Table 3
Moisture content, fresh (wet basis)	mean 58.2% (70.5% at the butt to 47.6% at the top)	not given	Table 1
Dry weight per hectare per year	4.9 t/ha/yr	5.0 t/ha/yr	Table 7
Calorific value per hectare per year	93 GJ/ha/yr	101 GJ/ha/yr	Table 7

- Bark has more ash, lignin and a higher calorific value than wood (Original Table 2).
- Under the Japan Woody Bioenergy Association (JWBA) standard, chips above 55% moisture are unsuitable as fuel, so drying is needed (Original, Discussion). By ash content, the wood is class 3 and the bark class 4 (Original, Discussion).

- Because of the high potassium, clinker (melted ash that hardens in the furnace) is a concern in combustion (Original, Discussion).

3. Authors' conclusions

- Kiri has the lowest calorific value per m³, but it grows fast, so its annual energy per hectare is almost the same as sugi's. With short rotations it may be usable for biofuel (Original, Abstract and Concluding Remarks).
- Wood higher in the stem has lower moisture. Using the lower stem for solid wood and the upper part for fuel could reduce drying cost (Original, Scenarios 2 and 3).

4. Points to keep in mind (Mitsue-it's view)

- The site was flat abandoned farmland with fertiliser and weeding, and no deer or steep-slope conditions. This is different from the hills of Mitsue.
- The per-hectare figures are estimates: 15 average trees multiplied by 1,055 trees/ha. The sugi comparator is a 48-year-old stand (note to Original Table 7).
- One site and one age (4 years) only.
- This summary is part of our research on planting kiri alone. It is not meant to argue against Yamauchi-san's plans.

Prepared by Mitsue-it (rob@mitsue.it), 15 October 2026.